An integral normalization of the Narayana-related recurrence

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Abstract

We prove Conjecture 8.7 of Amdeberhan, Moll and Vignat, in the stronger form that a positive integer initial value makes every term an integer. For a nonnegative integer μ, the value $2\,\mathrm{lcm}(1,\dots,2\mu)$ suffices, with the empty least common multiple interpreted as 1. The proof bounds each prime valuation of every coefficient in the defining quadratic recurrence.

Task 1: the source question and its resolution

The source's recurrence (7.10) is

$$a_n = \sum_{k=1}^{n-1} C_\mu(n,k)\,a_k a_{n-k}, \qquad C_\mu(n,k) = \frac{\binom{n+\mu-1}{n-k-1}\binom{n+\mu-1}{k-1}}{2\binom{n+\mu-1}{n-1}} \quad (n \ge 2). \tag{1}$$

The initial value a_1 may be chosen. In Definition 8.5, the source calls a rational sequence p-integral when every reduced denominator is a power of p. Conjecture 8.7 asks whether such an initial value and prime exist for every $\mu \in \mathbb{N}$.

Theorem 1. *Let $\mu \ge 0$ be an integer, and set*

$$L_\mu = \mathrm{lcm}(1,\dots,2\mu), \qquad D_\mu = 2L_\mu, \qquad L_0 = 1.$$

For every $n \ge 2$ and $1 \le k < n$, the number $D_\mu C_\mu(n,k)$ is a positive integer. Consequently, choosing $a_1 = D_\mu$ in (1) gives $a_n \in D_\mu \mathbb{Z}_{>0}$ for every $n \ge 1$. In particular, the sequence is p-integral for every prime p.

This is a nonzero positive construction. Denominator $1 = p^0$ is allowed by the stated definition; no restriction to the small experimental initial values is imposed in Conjecture 8.7.

A floor-function bound

Lemma 1. *For integers $t \ge 1$, $q \ge 2$ and $a, b \ge 0$, put*

$$\begin{aligned} F_q(a,b,t) = {} & \left\lfloor \frac{a+b+1}{q} \right\rfloor + \left\lfloor \frac{a+b+t}{q} \right\rfloor + \left\lfloor \frac{t-1}{q} \right\rfloor \\ & - \left\lfloor \frac{a}{q} \right\rfloor - \left\lfloor \frac{b}{q} \right\rfloor - \left\lfloor \frac{a+t}{q} \right\rfloor - \left\lfloor \frac{b+t}{q} \right\rfloor. \end{aligned}$$

Then $F_q(a,b,t) \ge -1$. Moreover, $F_q(a,b,t) \ge 0$ whenever $q \ge 2t-1$.

Proof. Write $a = Aq + r$, $b = Bq + s$ and $t = Tq + u$, where $0 \le r, s, u < q$. All integral multiples cancel, giving

$$F_q = \left\lfloor \frac{r+s+1}{q} \right\rfloor + \left\lfloor \frac{r+s+u}{q} \right\rfloor - \left\lfloor \frac{r+u}{q} \right\rfloor - \left\lfloor \frac{s+u}{q} \right\rfloor - \mathbf{1}_{u=0}. \tag{2}$$

If $u = 0$, the two subtracted floors vanish, so the result is at least -1. If $u > 0$, each subtracted floor is either 0 or 1. When both are 1, $r, s \ge q - u$, and hence $r + s + u \ge q$; thus even in this case the expression is at least -1. If at most one is 1, the same lower bound is immediate.

Suppose now $q \ge 2t - 1$. Since $q \ge 2$, we have $t < q$ and thus $u = t$. If neither subtracted floor in (2) is 1, the expression is nonnegative. If at least one is 1, the second positive floor is at least 1. If both are 1, then $r + s + 1 \ge 2q - 2t + 1 \ge q$, so the first positive floor is also at least 1. This proves nonnegativity in all cases. □

The denominator bound and induction

Proof of Theorem 1. Set $a = k-1$, $b = n-k-1$, and $t = \mu+1$. Expanding the binomial coefficients gives the positive rational number

$$2C_\mu(n,k) = \frac{(t-1)!(a+b+1)!(a+b+t)!}{a!\,b!\,(a+t)!\,(b+t)!}. \tag{3}$$

For a prime ℓ, Legendre's formula gives

$$v_\ell\big(2C_\mu(n,k)\big) = \sum_{e \ge 1} F_{\ell^e}(a,b,t).$$

The sum is finite. By Lemma 1, all terms with $\ell^e \ge 2\mu + 1$ are nonnegative, and each remaining term is at least -1. Therefore

$$v_\ell\big(2C_\mu(n,k)\big) \ge -\#\{e \ge 1 : \ell^e \le 2\mu\} = -v_\ell(L_\mu).$$

For $\mu = 0$ the set is empty and $L_0 = 1$. Thus every prime valuation of $2L_\mu C_\mu(n,k) = D_\mu C_\mu(n,k)$ is nonnegative, proving that this positive rational number is an integer.

Finally put $b_n = a_n/D_\mu$. The initial value is $b_1 = 1$, and (1) becomes

$$b_n = \sum_{k=1}^{n-1} \big(D_\mu C_\mu(n,k)\big) b_k b_{n-k}.$$

Induction proves $b_n \in \mathbb{Z}_{>0}$ for every n, as required. □

Verification and source scope

The proof is uniform in all indices and does not rely on finite testing. The accompanying standard-library Python verifier checks 153,450 exact recurrence coefficients, 9,300 factorial-ratio identities, 13,533,960 floor-function cases, 840 sequence/scaling identities, and 21 terms from the source's examples. All pass. The whole current arXiv manuscript and the 25-page journal article were inspected. The latter retains Conjecture 8.7 on page 22. Its postscript resolves the separate logconvexity Conjecture 9.1 and does not assert a resolution of Conjecture 8.7. This note was reviewed by its author agent; no independent peer review is claimed. Dated status evidence and bounded literature-search limits are recorded in the accompanying audit files.

Four follow-up problems

2. **Task 2.** Determine the least positive integer M_μ such that $M_\mu C_\mu(n,k)$ is integral for all $n \geq 2$, $1 \leq k < n$, and give its prime factorization as a function of μ.

3. **Task 3.** Determine the least positive integer initial value A_μ for which the entire sequence (1) is integral. Compare A_μ with M_μ; cancellation within the recurrence is allowed.

4. **Task 4.** For the canonical initial value $a_1 = 2$, classify the primes occurring in reduced denominators and determine $\liminf_{n\to\infty} v_\ell(a_n)/n$ for each such prime ℓ.

5. **Task 5.** Classify rational parameters $\mu > -1/2$ for which a nonzero rational initial value makes every term of (1) an integer, using generalized binomial coefficients in that recurrence.

References

[1] T. Amdeberhan, V. H. Moll and C. Vignat, *A probabilistic interpretation of a sequence related to Narayana polynomials*, Online Journal of Analytic Combinatorics 8 (2013), paper 1, 1–25. doi:10.61091/ojac-801. Preprint arXiv:1202.1203.

[2] MathDB 322992, *The p-integrality conjecture for the Narayana-related sequence.*